

Pratham Dwivedi

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PROFILE

Computer Science Undergraduate specializing in machine learning systems design and automated infrastructure. Proficient in transforming procedural ML code into configuration-driven, modular OOP architectures. Hands-on experience building reproducible data pipelines with DVC and AWS S3, managing unified experiment registries via MLflow, and orchestrating continuous automation workflows with GitHub Actions.

EDUCATION

Indian Institute of Information technology, Nagpur

Bachelor of Technology
Cumulative GPA: 9.10

Nagpur

August 2024 - Present

PROJECT EXPERIENCE

Kidney Tumor detection - Python, TensorFlow, FastAPI, Docker, DVC, MLflow, GitHub Actions

- Trained EfficientNetB0 on CT Kidney Dataset (12,446 images); achieved 97.9% accuracy, 94.4% F1, and 0.998 ROC-AUC.
- Quantized model to 4.6MB TFLite artifact eliminating 1.2GB TF overhead; deployed containerized FastAPI service with async lifespan loading.
- Engineered dual-stage GitHub Actions CI/CD pipeline with pytest suites validating API endpoints before Docker image push.

Repo : <https://github.com/prathamd69/kidney-tumor-detection>

CiFix - Civic Issue Routing System | Python, HuggingFace Transformers, BLIP, RoBERTa

- Built BLIP + SentenceTransformer multimodal pipeline auto-rejecting image-text mismatches below configurable cosine similarity threshold.
- Deployed RoBERTa-large-MNLI zero-shot classifier routing complaints to departments, with confidence-gap logic flagging edge cases.

Repo: <https://github.com/prathamd69/decorator>

PyDeepFlow | Open-Source Deep Learning Library - Python / NumPy

- Implemented 8 weight initialization strategies in NumPy, improving ANN convergence without manual user configuration.
- Merged PR #59 to main after 8 commits and peer review, adding flexible per-layer init API (string/list/auto modes).

PR: github.com/ravin-d-27/PyDeepFlow/pull/59

SKILLS

- Programming & Libraries: Python, C, NumPy, Pandas, TensorFlow/Keras, scikit-learn
- Tools & Frameworks: Git/GitHub, Flask, FastAPI, DVC, MLflow, Docker, GitHub Actions (CI/CD)
- Areas: Machine Learning, MLOps, Deep Learning, Computer Vision, NLP